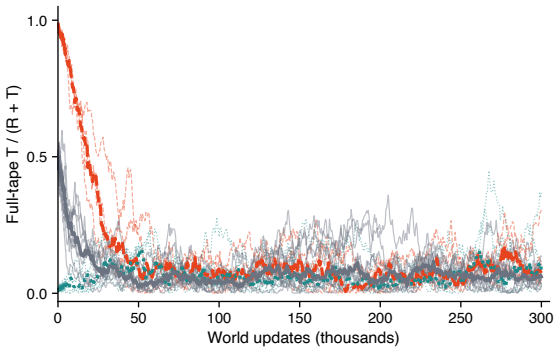
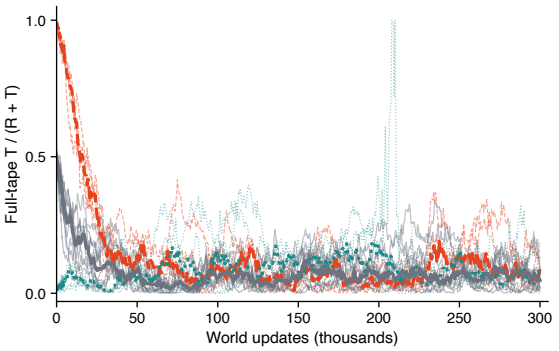
a **Historical / benign: L32, mutation on**



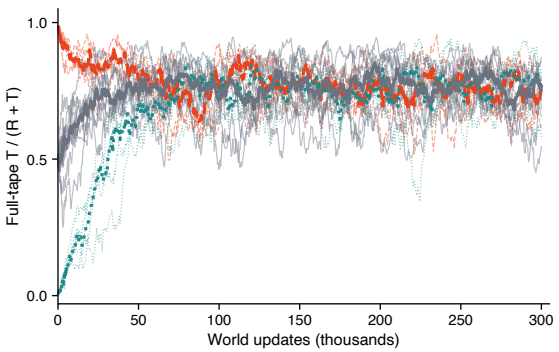
Thin worlds; thick medians; solid: 50:50 (10 worlds)
Dashed: R at 1% into T (5); dotted: T at 1% into R (5)
R partial-copy / T full-tape; C1 corrected Z80 / historical C0

b **C1 / benign: L32, mutation on**



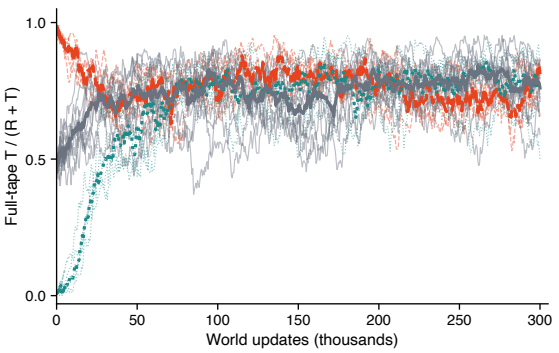
Thin worlds; thick medians; solid: 50:50 (10 worlds)
Dashed: R at 1% into T (5); dotted: T at 1% into R (5)
R partial-copy / T full-tape; C1 corrected Z80 / historical C0

c **Historical / lethal: L32, mutation on**



Thin worlds; thick medians; solid: 50:50 (10 worlds)
Dashed: R at 1% into T (5); dotted: T at 1% into R (5)
R partial-copy / T full-tape; C1 corrected Z80 / historical C0

d **C1 / lethal: L32, mutation on**



Thin worlds; thick medians; solid: 50:50 (10 worlds)
Dashed: R at 1% into T (5); dotted: T at 1% into R (5)
R partial-copy / T full-tape; C1 corrected Z80 / historical C0

e **Complete environmental endpoint ledger: 110 worlds per implementation**

L bytes	Zero rule	Mutation	Start	n C1/H	T C1	T H	Core C1	Core H	T tails C1	T tails H
16	benign	off	50:50	10/10	0.62	0.70	1.00	1.00	1	3
16	benign	on	R at 1%	5/5	0.50	0.33	0.04	0.03	70	56
16	benign	on	T at 1%	5/5	0.58	0.88	0.01	0.04	46	155
16	benign	on	50:50	10/10	0.70	0.52	0.05	0.05	156	134
16	lethal	off	50:50	10/10	0.63	0.32	1.00	1.00	2	2
16	lethal	on	R at 1%	5/5	0.81	0.87	0.72	0.73	3305	3476
16	lethal	on	T at 1%	5/5	0.81	0.81	0.74	0.00	3307	0
16	lethal	on	50:50	10/10	0.86	0.86	0.72	0.72	3234	3336
32	benign	off	50:50	10/10	0.28	0.25	1.00	1.00	1	2
32	benign	on	R at 1%	5/5	0.09	0.07	0.88	0.80	448	391
32	benign	on	T at 1%	5/5	0.08*	0.08	0.76	0.87	162	396
32	benign	on	50:50	10/10	0.05	0.06	0.85	0.84	229	308
32	lethal	on	R at 1%	5/5	0.81	0.76	0.91	0.88	4477	3879
32	lethal	on	T at 1%	5/5	0.77	0.78	0.88	0.93	4000	4272
32	lethal	on	50:50	10/10	0.76	0.77	0.90	0.92	4046	4135

World medians; H = historical C0. Benign zero executes; lethal zero halts. Mutation off: 100k; on: 300k updates.
T = T/(R+T); core = 5E ED B0-bearing cells /20,000; T tails = distinct sequences at bytes 4–(L–1) per world.
*C1 L32 benign T-at-1%: T median uses 4/5 ratios; core/tail medians use all 5 worlds, including the zero world.

f **EXPLORATORY continuation and host controls; registered stability verdict**

Source soup	Execution zero rule	Worlds	Mean T fraction, start	Mean T fraction, end
benign	benign	10	0.071	0.070
benign	lethal	10	0.071	0.110
lethal	benign	10	0.734	0.685
lethal	lethal	10	0.733	0.740

Continuation: historical Z80 L32; mean T/(R+T), end = 2,000 updates; both classes pre-exist.

Historical lethal L32 hosts: fraction unchanged native 0.0137, detox 0.0128; n=400×64 encounters.

Detox replaces payload 0x00/0x76 with random non-toxic bytes; host is partner B of soup executors.

E1, corrected C1 benign mutation-off: ≥7/10 with T/(R+T) 0.3–0.7; L16: 4/10, L32: 4/10; failed.

Continuation and host tests are distinct from registered competitions; variable endpoints do not establish neutral drift.